Outliers – Unusual Values in Y

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Identifying Residual "Outliers"

- Residuals are the difference between the actual DATA and our MODEL's predictions:
  - $e_i = y_i - \hat{y}_i$

- Problems with the "raw" residuals as a measure of error:
  - **Scale**: Unless we standardize the residuals in some way, larger scales are always going to look like they more errorful than smaller scales.

  - **Inclusion**: If an extreme value is included to begin with, it is going to inflate measures of variability. Paradoxically, this will make the value appear less extreme.

Identifying Residual "Outliers"

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> DAT2
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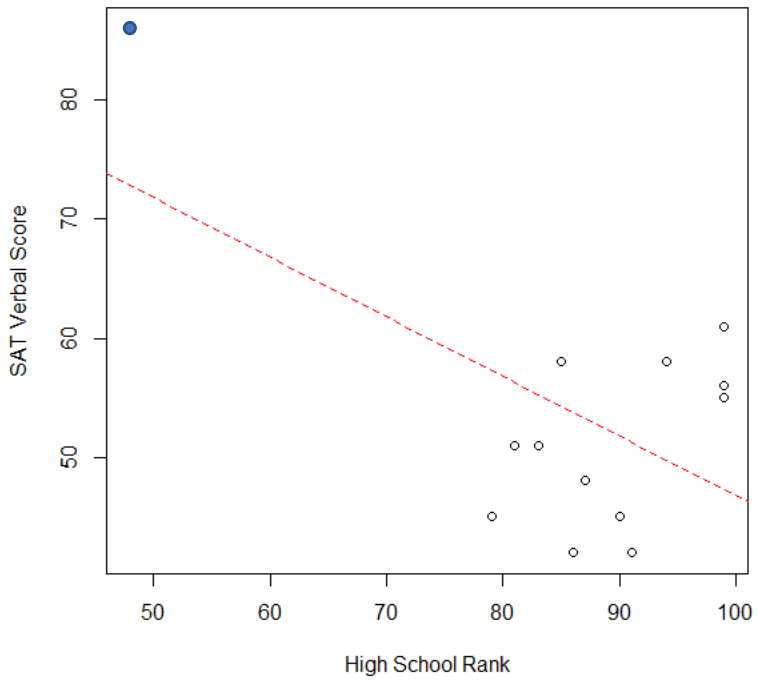
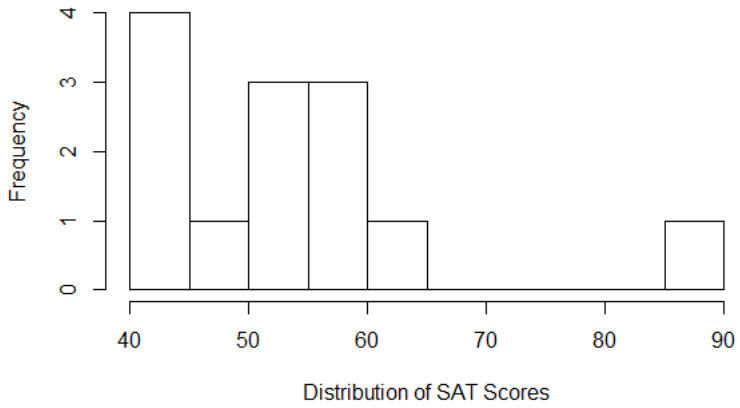
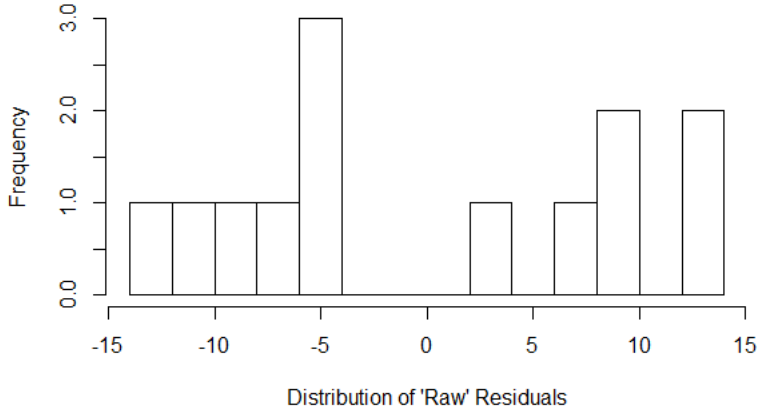
ID	SAT	HSR
1	42	91
2	48	87
3	58	85
4	45	79
5	45	90
6	86	48
7	51	83
8	56	99
9	51	81
10	58	94
11	42	86
12	55	99
13	61	99







R uses Standardized Deleted Residuals by Default

- Note that the book used *studentized* residuals rather than standardized residuals, but conceptually these two calculations obtain a similar result:

- Standardized residuals: $e_{s,d_i} = \frac{(\mathrm{y_i} - \hat{y}_i)}{s * \sqrt{1 - h_i}}$

- The standardized deleted residual solves both of our problems:
  - **Scale**: The residuals are "standardized" to the standard deviation of all the residuals.
    - This is where *studentized* and *standardized* calculations differ. Studentized takes this adjustment a step further and calculates 's' excluding the current residual as well.

  - **Inclusion**: We reduce the influence of a single variable by accounting for it's leverage.

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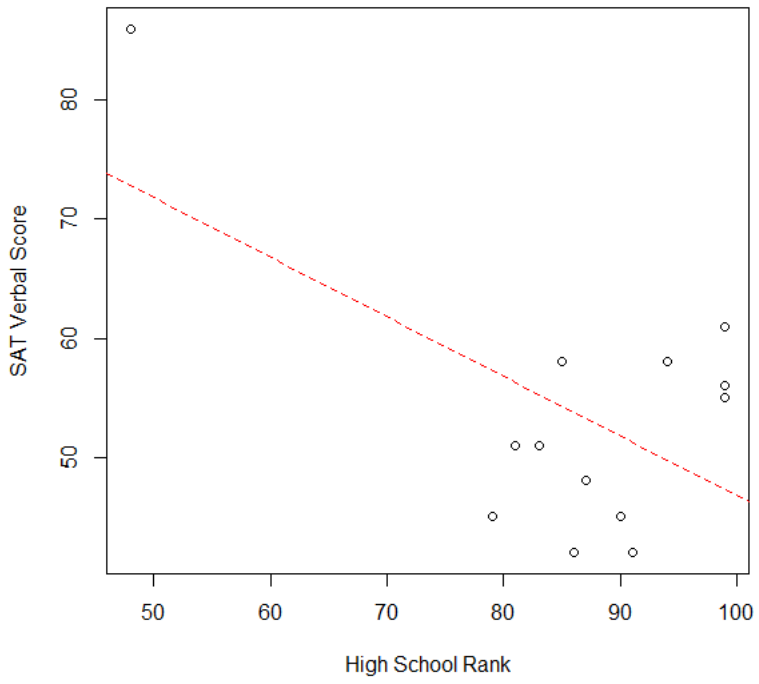
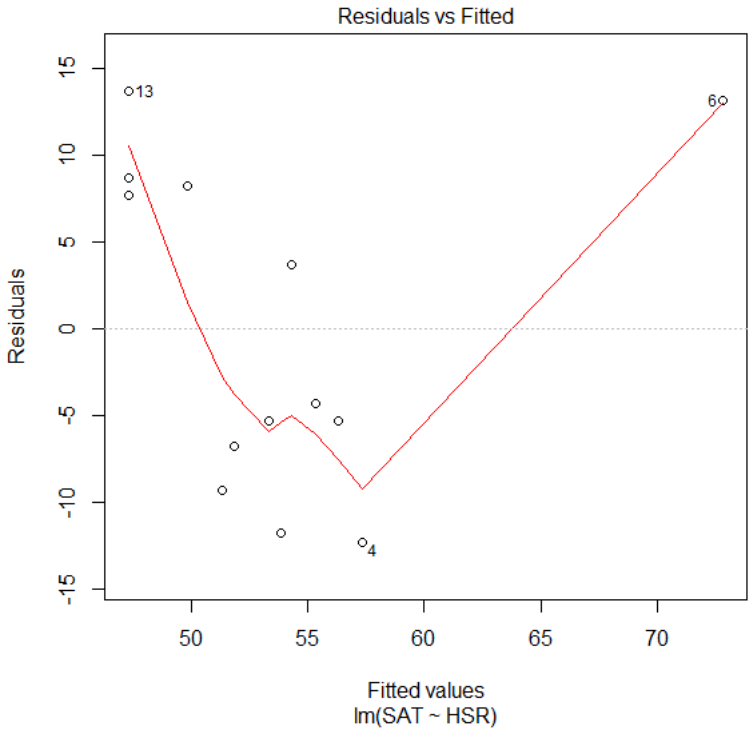
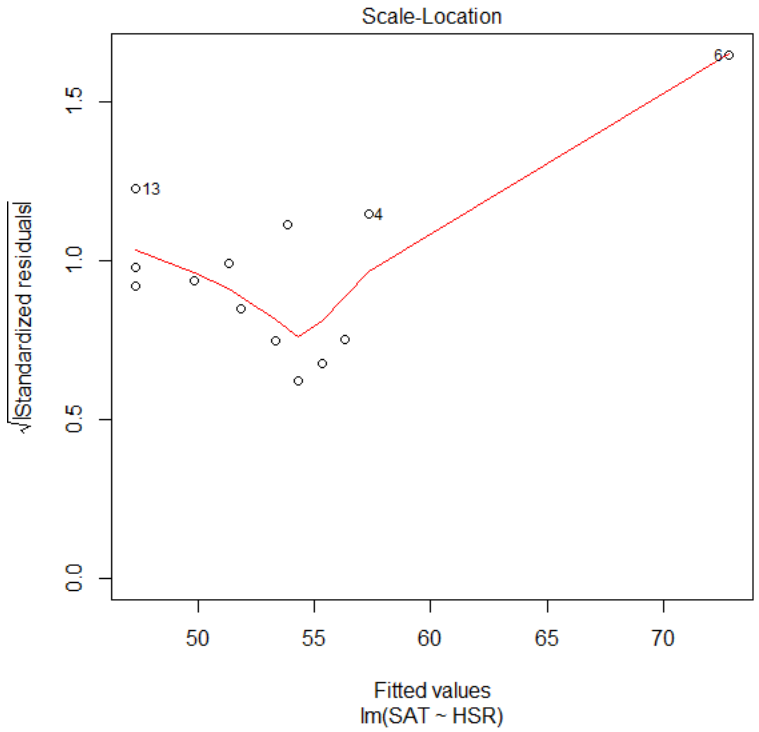
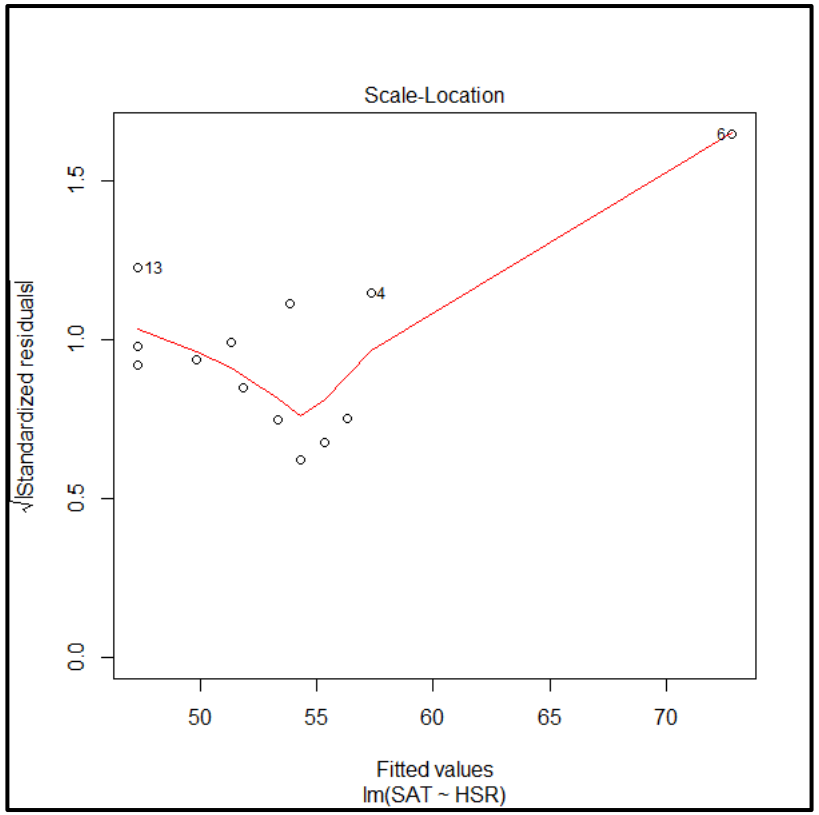
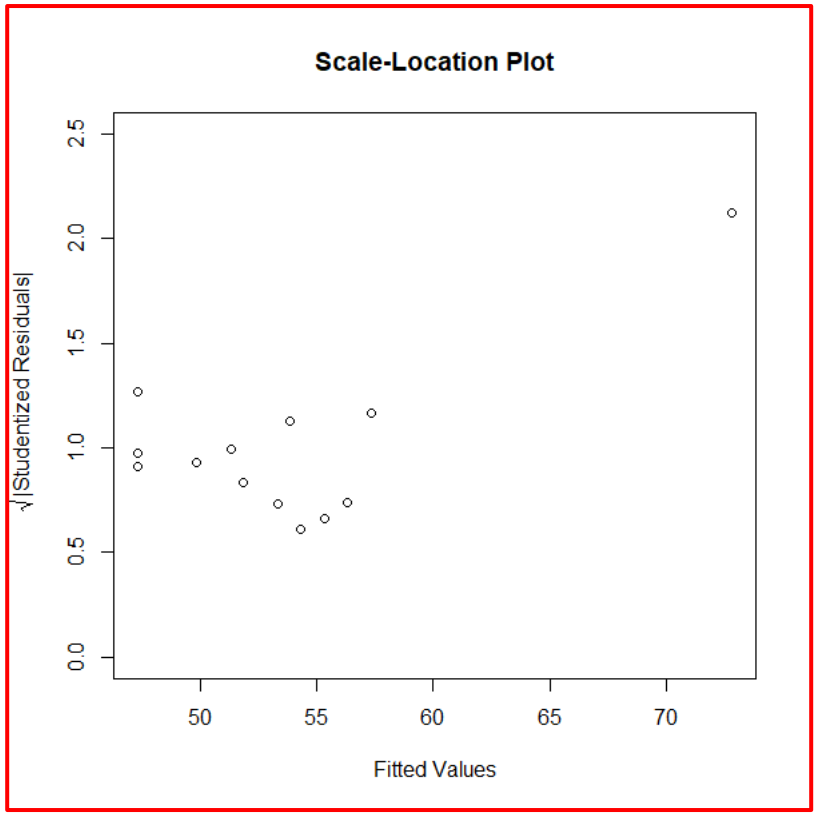






Standardized versus Studentized Residuals





$$e_{s,d_i} = \frac{(\mathrm{y_i} - \hat{y}_i)}{s * \sqrt{1 - h_i}}$$

$$e_{s,d_i} = \frac{(\mathrm{y_i} - \hat{y}_i)}{s_{(i)} * \sqrt{1 - h_i}}$$

Why are outliers bad?

- Ideally, we want a model that accurately represents all of the data.

- If a single (or several data points) are large outliers, this could be diagnostic of several things:
  - Some error has occurred in the coding/recording of the data,
  - That data point does not actually belong to the sample in question,
  - That data point could be pulling our model away from the other data points.

- Thus, the presence of an outlier is not grounds for exclusion of data in itself, but it should raise questions that force us to take a harder look at our data and our model.